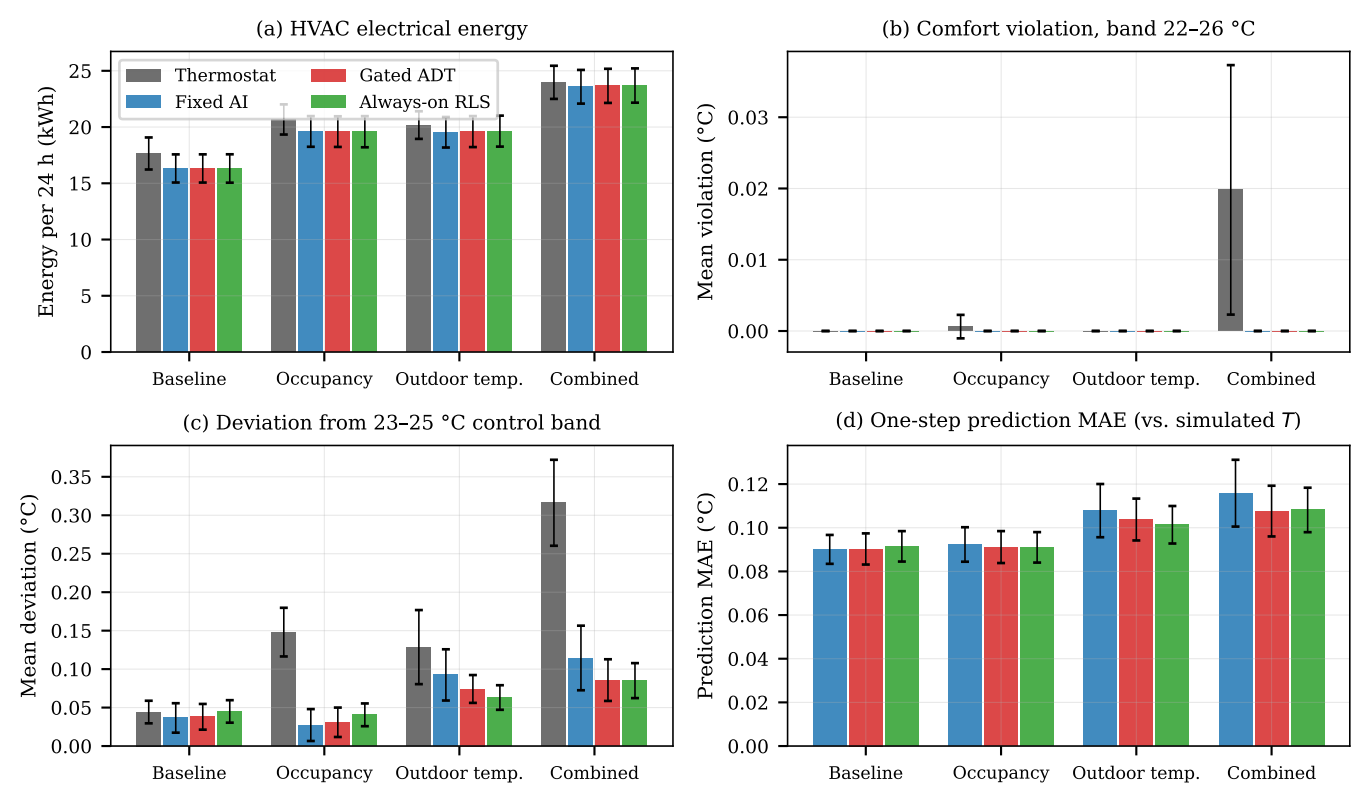
(a) HVAC electrical energy
Thermostat
Fixed AI
Gated ADT
Always-on RLS
Energy per 24 h (kWh)
0
5
10
15
20
25
Baseline
Occupancy
Outdoor temp.
Combined
(b) Comfort violation, band 22–26 °C
Mean violation (°C)
0.00
0.01
0.02
0.03
Baseline
Occupancy
Outdoor temp.
Combined
(c) Deviation from 23–25 °C control band
Mean deviation (°C)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Baseline
Occupancy
Outdoor temp.
Combined
(d) One-step prediction MAE (vs. simulated T)
Prediction MAE (°C)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Baseline
Occupancy
Outdoor temp.
Combined